**Reading Chapter 1 - Computational Thinking and Problem Solving "The quest of THE ALGORITHM"**
https://docs.google.com/document/d/1p6_7n5m1YwSYE5g2kaxcPp_9_TlNSVmkWYAjaznq0c8/

# Week 1: Tasks Lab 1 - Tasks

# Basic Logic Building

**THE HINTs TO ALL THE PROBLEMs are given in Chapter 1's end. I recommend that if you are stuck on any problem then you should look into the exercise section and see the hints to the problems.**

**Problem 1**

Let's suppose that we have a balance and metal balls which are equal in weights except one which is heavier than all. Now we have to tell which one is the heaviest ball? The only way you can compare the weight of the two balls is placing on the balance and checking which one bends down.

**Part1**: Solve the above problem for **18 and 27 balls in just 3 tries.**
**Part2**: Solve the same problem for **50 and 81 balls in just 4 tries.**

**Problem 2**

You have two drums of capacity of 8 litres and 5 litres and a resource of Oil. Can you use these two drums to serve 1 litre, 2 litre, 3 litre, 4 litre, and 6 litres and 7 litres to any customer. Note that there is no mark of litres on the drums.
**BONUS:** Now you have to measure again to answer the same question, but this time, you have a 9-litre and 6-litre drum. Can you solve that same problem? If yes, how? If not, why not?

**Problem 3**

On your first day at university, the teacher suggested that it would be a good idea for each student to meet every other student in the class. The teacher said, "When you meet, please shake hands and introduce yourself by name." If there were 20 students in the class, how many total handshakes were there? What about if there are N students can you tell in terms of N what is the total count of shake-hands.

**Problem 4**

If you were to construct a 8 x 8 chess board, how many squares (within that chess board) would there be in total? (64 is not the answer).
You remember the 8 balls problem, now you have N balls you have to tell how many times you have to weigh at max to figure out which ball is the heavier one?
Just a direction: First try to figure out, 9, 16, 21, 27, 30, 50, 81 and then you will see what the pattern is!

**Problem 5**

You are given 2 eggs. You have access to a 100-story building. Eggs can be very hard or very fragile means it may break if dropped from the first floor or may not even break if dropped from the 20th floor. Both eggs are identical. You need to figure out the highest floor of a 20-story building from where an egg can be dropped without breaking. The question is how many drops you need to make. You are allowed to break 2 eggs in the process.

Hint: Binary Search will not work here.

**Problem 6**
Divide $277 (in whole $ increments) into a number of bags so that I can ask for any amount between $1 and $277, and you can give me the proper amount by giving me a certain number of these bags without opening them. What is the minimum number of bags you will require?

**Problem 7**
A corporate businessman has two cubes on his office desk. Every day he arranges both cubes so that the front faces show the current day of the month. What numbers are on the faces of the cubes to allow this?
Note: You can't represent the day "7" with a single cube with a side that says 7 on it. You have to use both cubes all the time. So the 7th day would be "07".

**Problem 8**
A pot contains 75 white beans and 150 black ones. Next to the pot is a large pile of black beans. A somewhat crazy cook removes the beans from the pot, one at a time, according to the following strange rule: He removes two beans from the pot at random. If at least one of the beans is black, he places it on the bean-pile and drops the other bean, no matter what color, back in the pot. If both beans are white, on the other hand, he discards both of them and removes one black bean from the pile and drops it in the pot. At each turn of this procedure, the pot has one less bean in it. Eventually, just one bean is left in the pot. What color is it?

**Problem 9**
So you have an infinite plane and lines, lines can intersect each other to make regions, e.g. 1 line will produce to regions, 2 lines 4. Can you tell with argument at maximum how many regions will be there if you have n lines? And why? (Any answer without why will be unacceptable).

**Problem 10**
You have two strings whose only known property is that when you light one either end of either string it takes exactly one hour to burn. The rate at which the strings will burn is completely random and each string is different. How do you measure 1.5 hours and 45 minutes?

---

**Problem 11**
There are four numbers: 1, 2, 5, 10 that you have to carry in a boat across a river. Every time you may not carry more than two numbers , and the time of traveling is associated with the largest of two numbers. One number has to be in the boat during the return journey, which also defines the time for this trip. What is the minimum total time to carry all numbers from one bank to another?

**Problem 12**

**You wake up after your Titanic sank, and you find yourself on the shore of the sea, an island. Beside everything, what an island has (its natural beauty), there are tribesmen there too. You come to know that every tribesman of this island is either faker(he may or may not lie) or trustworthy(always say the truth). You can ask them questions like who is a faker. And who is trustworthy? There are a total of 100 people on this island. And you can ask as many questions as you want. The only information known is that the number of Trustworthiness is greater than fakers. Any answer having greater than 500 questions is not worthy enough to be considered. Figure out who is a faker and who is a trustworthy person? You can ask questions from any person A about person B like: Is B a trustworthy person?**

**Problem 13**

- A duck is pursued by a fox, escapes to the center of a perfectly circular pond. The fox cannot swim and the duck cannot take flight from the water. The fox is 4 times faster than the duck. Assuming the fox and duck pursue optimal strategy, is it possible for the duck to reach the edge of the pond and fly away without being eaten? If so how? (You can find its answer on the Internet very easily, you can use Internet but I would say first try for it and then go on finding it) [2]
- On what parameters (relation between their speeds) fox will always be able to eat it, and on what factors duck will always escape.

**Problem 14**

Let us play a game. This is a One-Dimensional game in which we have many people/life(**N** people) in a single row. Accidently these people/life are affected by a deadly virus. This virus spreads people to people one by one in specific order and maybe when we start the game then there are already many people/life affected by the virus but fortunately Dr. Sarim made the antidote for that virus. As we know, viruses affect people continuously. So, when we check any life for the virus affected on that life or not, then it will take a little bit of time but during that time the virus will affect another life who will be next to previous affected life. We can check and give an antidote to people/life one by one, meaning only one life will be checked and get antidote at a time. So, we need to save the maximum people from the virus. How many minimum people will be affected by the virus during finding last affected by the virus ?

```
O===O
 \O/
  |
 /\
```

**Thinking in power...!**

**Good luck**

**Reading Chapter 1 - Computational Thinking and Problem Solving "The quest of THE ALGORITHM"**

https://docs.google.com/document/d/1p6_7n5m1YwSYE5g2kaxcPp_9_TlNSVmkWYAjaznq0c8/